Theoretical Foundations: Maritimization Theory and Programmatic Systematization

2.1 Introduction: Grounding Blue Sociology in Maritime Sociology Traditions

The development of Blue Sociology as a complementary renewal of maritime sociology requires explicit engagement with the field's foundational theoretical and programmatic contributions. While contemporary oceanic turn scholarship has introduced vital conceptual innovations—hydrosocial relations, more-than-human ontologies, infrastructural politics—these insights risk remaining disconnected from the empirical and analytical traditions that have long examined sea-society relations. Two foundational contributions stand out as particularly generative for Blue Sociology's theoretical architecture: Ludwik Janiszewski's maritimization theory, developed in postwar Poland, and Michael Poole's programmatic systematization of maritime sociology as a coherent field of inquiry.

These two frameworks, though emerging from distinct intellectual contexts and addressing different analytical problems, provide complementary foundations for Blue Sociology. Janiszewski offers an ontological framework for understanding how marine environments and human societies become mutually constitutive through processes of maritimization (*marynizacja*), while Poole provides methodological and thematic systematization that links substantive maritime topics to general sociological theory. Together, they ground Blue Sociology's dual focus on material sea-society entanglements and the social organization of maritime labor, institutions, and power.

This section provides comprehensive coverage of both theoretical traditions, examining their historical contexts, core concepts, empirical applications, and contemporary relevance for Blue Economy analysis. We demonstrate how Janiszewski's semi-deterministic framework illuminates the Marine-Maritime-Oceanic tripartite distinction that structures Blue Sociology, while Poole's programmatic agenda supports the framework's six core analytical dimensions. The synthesis reveals how these mid-to-late twentieth-century contributions remain essential for addressing twenty-first-century challenges of blue growth, ocean governance, and social sustainability.

2.2 Ludwik Janiszewski's Maritimization Theory: Ontological Foundations of Sea-Society Entanglement

2.2.1 Historical and Intellectual Context

Ludwik Janiszewski developed his maritimization theory within the specific institutional and political context of postwar Poland, a period marked by dramatic territorial reconfiguration, socialist state-building, and the emergence of maritime sociology as a distinct subdiscipline. Poland's westward territorial shift following World War II incorporated a substantial Baltic coastline and major port cities—Gdańsk, Szczecin, and Gdynia—transforming a historically land-oriented nation into a maritime state [1]. This geopolitical transformation created both practical imperatives for understanding coastal development and intellectual opportunities for sociological inquiry into sea-society relations.

The institutional context for Janiszewski's work was shaped by the establishment of coastal academic centers and research institutes dedicated to maritime studies. Universities in Gdańsk and Szczecin emerged as principal sites for maritime sociology, supported by regional institutes that promoted maritime education, workforce development, and policy research [2]. This institutional infrastructure enabled sustained empirical investigation of fishing communities, seafarers, port workers, and shipbuilding sectors—the empirical foundations upon which Janiszewski built his theoretical framework [3].

Janiszewski's intellectual development occurred during the 1960s and 1970s, a period when Polish sociology was navigating complex relationships between Marxist theoretical frameworks, empirical social research, and practical policy engagement. While the supplied sources document the institutional growth of maritime studies in postwar Poland, they provide limited detail on the specific political and ideological pressures that shaped Janiszewski's theoretical choices. What is clear is that his approach combined structural analysis of maritime economic systems with attention to human agency, cultural practices, and identity formation—a balance that distinguished his work from more deterministic materialist frameworks [1].

The publication of *Rybacy dalekomorscy: Studium socjologiczne* (Distant-Water Fishermen: A Sociological Study) in 1967 marked a key moment in Janiszewski's empirical and theoretical development [4]. This monograph provided detailed sociological analysis of Poland's first generation of distant-water fishermen on the Western Pomeranian coast, examining occupational identity, family life, fleet organization, and the social-historical genesis of this new maritime workforce. The study exemplified Janiszewski's methodological approach: combining historical analysis, structural description of economic and institutional arrangements, and qualitative investigation of workers' lived experiences and subjective orientations toward their occupation [4].

2.2.2 Core Theoretical Concepts: Marynizacja and the Two-Factor Framework

At the heart of Janiszewski's contribution is the concept of *marynizacja* (maritimization), which denotes the penetration and shaping of a society or social group by the sea and maritime activity. Maritimization encompasses the processes through which maritime conditions, technologies, and institutions influence economic organization, social roles, cultural practices, mentalities, and local identities [1]. The concept provides a heuristic for analyzing how coastal and maritime communities are constituted through ongoing interaction between environmental-material conditions and human social organization.

Janiszewski's framework is structured around two interdependent factors that jointly produce maritimization: the marine factor (*czynnik morski*) and the human factor (*czynnik ludzki*). This two-factor structure distinguishes his approach from both environmental determinism and purely constructivist accounts of social organization [1].

The Marine Factor encompasses the sea's material, technological, and systemic influences on social life. This includes:

- Environmental conditions: oceanographic characteristics, weather patterns, resource distributions, and ecological dynamics that structure maritime activities and create specific risks, opportunities, and constraints [1].

- Maritime technologies and infrastructure: vessels, navigation systems, port facilities, processing plants, and communication technologies that mediate human engagement with marine environments [1].

- Institutional arrangements: property regimes, regulatory frameworks, fleet organizations, market structures, and state policies that organize maritime economic activity and connect localities to broader maritime flows [1].

- Spatial configurations: the physical geography of coastlines, harbors, and maritime routes that shape patterns of settlement, trade, and territorial control [1].

The marine factor represents the structural ground of maritimization—the relatively durable material and institutional conditions that orient social possibilities and constrain developmental pathways. However, Janiszewski explicitly rejects environmental determinism: the sea does not mechanically determine social outcomes but rather creates predispositions, opportunities, and constraints that human actors interpret and respond to [1].

The Human Factor comprises the actors, practices, and social processes through which maritime influences are appropriated, mediated, and transformed. This includes:

- Maritime occupational groups: seafarers, fishers, port workers, shipbuilders, and maritime professionals whose labor directly engages marine environments and whose identities are shaped by maritime work [1], [4].

- Local elites and planners: state officials, port authorities, fleet managers, and regional development agencies that make strategic decisions about maritime investment, infrastructure, and workforce development [1].

- Cultural practices and identity formations: maritime traditions, occupational cultures, family patterns, educational practices, and collective identities that reproduce maritime orientations across generations [1], [5].

- Social organizations and institutions: unions, cooperatives, professional associations, educational institutions, and community organizations that mediate between individual maritime workers and broader economic and political structures [1].

The human factor represents agency, interpretation, and social organization—the ways in which communities and groups actively shape their relationships with maritime environments rather than passively receiving environmental influences. Janiszewski emphasizes that maritime development outcomes reflect the interplay of structural maritime forces and local capacity to organize labor, institutions, and cultural practices around maritime opportunities [1].

2.2.3 Semi-Deterministic Framework and Mechanisms of Entanglement

Janiszewski's approach is characterized as semi-deterministic: environmental and infrastructural forces orient social possibilities while social actors and institutions translate those orientations into particular developmental outcomes [1]. This theoretical stance navigates between structural determinism and voluntarism, recognizing both the constraining power of material conditions and the transformative capacity of human agency and social organization.

The theory maps several key mechanisms of sea-society entanglement through which maritimization produces durable social forms:

Economic linkage mechanisms involve the creation of maritime occupations, firms, and industries that reconfigure local labor markets and economic structures. When fishing fleets, shipping companies, port operations, or shipbuilding facilities are established in a coastal region, they generate employment opportunities, income flows, and economic dependencies that orient local development toward maritime sectors [1]. These economic linkages create material interests in maritime activity that shape political coalitions, investment priorities, and regional identities.

Institutional embedding mechanisms involve the establishment of port authorities, maritime educational institutions, fleet management organizations, and regulatory agencies that lock in maritime developmental pathways [1]. Once substantial institutional infrastructure is created to support maritime activity, path dependencies emerge: institutional actors develop vested interests, specialized knowledge accumulates, and switching costs for alternative development strategies increase. Maritime institutions thus become self-reinforcing, reproducing maritime orientations even as specific economic conditions change.

Cultural transmission mechanisms involve the reproduction of maritime identities, practices, and orientations through families, workplaces, schools, and community organizations [1], [5]. When children grow up in households where parents work at sea, when schools incorporate maritime education and training, when local cultural practices celebrate maritime heritage, maritime identities become embedded in social reproduction. These cultural mechanisms create subjective orientations toward maritime life that complement economic and institutional structures.

Spatial concentration mechanisms involve the clustering of maritime activities, populations, and infrastructure in specific coastal locations, creating agglomeration effects and distinctive maritime urban forms [1]. Port cities develop specialized labor markets, service sectors, and built environments adapted to maritime functions. This spatial concentration reinforces maritimization by creating dense networks of maritime actors and institutions in specific places.

Janiszewski's framework thus conceptualizes maritimization as a process of mutual constitution: the sea shapes societies through material constraints and opportunities, while societies shape their relationships with the sea through institutional choices, cultural practices, and technological mediations. This dialectical understanding anticipates contemporary relational ontologies in oceanic turn scholarship while remaining grounded in empirical analysis of specific maritime communities and sectors [1].

2.2.4 Empirical Applications: From Fishermen to Regional Development

Janiszewski's maritimization framework was developed through and applied to diverse empirical domains, demonstrating its analytical versatility across scales and sectors.

Fishing communities constituted Janiszewski's primary empirical focus. His 1967 monograph on distant-water fishermen provided detailed analysis of how Poland's postwar expansion of industrial fishing created a new occupational community [4]. The study examined the social-historical genesis of this workforce, analyzing recruitment patterns, occupational socialization, family adaptations to prolonged absences at sea, and workers' subjective orientations toward their profession. Janiszewski documented how the marine factor—the establishment of modern trawler fleets, distant-water fishing grounds, and processing infrastructure—interacted with the human factor—workers' motivations, family strategies, fleet policies, and occupational cultures—to produce a distinctive maritime social formation [4].

Coastal communities and maritime identity have been analyzed using Janiszewski's framework to understand how coastal populations develop and reproduce maritime orientations. Research in the Polish maritime sociology tradition has examined "maritime upbringing" (*wychowanie morskie*)—deliberate educational and cultural programs designed to foster maritime identities, knowledge, and career orientations among coastal youth [5]. These studies reveal how institutional actors (schools, scouting organizations, maritime leagues) actively work to reproduce maritimization through cultural transmission, demonstrating the human factor's role in sustaining maritime social formations across generations [5].

Port cities have been analyzed through the maritimization lens to understand urban transformations driven by changing maritime governance and economic structures. Researchers have used the concept to examine how port cities experience phases of intensified maritimization—when maritime commerce, institutions, and cultures dominate urban life—and subsequent "demaritimization" when ports decline, maritime employment contracts, and continental influences penetrate urban governance [6]. This application reveals maritimization as a dynamic, historically variable process rather than a static condition.

Maritime industries including shipbuilding have been studied using maritime sociological frameworks that inherit Janiszewski's attention to organizational change, labor relations, and institutional dynamics [7], [8]. Research on Polish shipyards during post-socialist transformation has examined how global competitive pressures, ownership restructuring, and technological change reshape workplace organization and worker identities in maritime industrial sectors [8]. These studies demonstrate the framework's utility for analyzing how global economic forces interact with local institutional and cultural contexts to produce specific trajectories of maritime industrial change.

Regional development has been analyzed using maritimization to interpret how regional policies, university centers, and development institutes mediate the integration of maritime economies into broader regional strategies [2], [7]. Studies have examined how coastal regions in Poland—particularly Gdańsk and Szczecin—developed distinctive maritime development models shaped by the interplay of maritime economic opportunities, institutional capacity, and regional political coalitions [2]. This application scales Janiszewski's framework from occupational communities to regional political economy.

2.2.5 Polish Maritime Sociology Tradition and Institutional Context

Janiszewski's work seeded a distinctive Polish maritime sociology tradition that has persisted as an institutionalized subdiscipline. This tradition is characterized by several features:

Institutional concentration in coastal academic centers, particularly universities in Gdańsk and Szczecin, which developed specialized research programs, degree offerings, and faculty positions focused on maritime sociology [2], [7]. These institutions created stable organizational bases for sustained empirical research and theoretical development, enabling cumulative knowledge production across generations of scholars.

Interdisciplinary orientation linking sociology with maritime economics, law, regional studies, and policy analysis [2], [7]. Polish maritime sociology has maintained productive relationships with adjacent disciplines, contributing to integrated analyses of maritime development that combine social, economic, legal, and spatial dimensions.

Empirical grounding in detailed studies of specific maritime occupational groups, communities, and industries [4], [8]. The tradition has emphasized fieldwork, qualitative methods, and close engagement with maritime workers and institutions, producing rich descriptive accounts alongside theoretical development.

Policy engagement with regional development agencies, maritime education institutions, and industry organizations [2], [5]. Polish maritime sociologists have actively participated in policy debates about maritime workforce development, regional planning, and maritime cultural programs, maintaining connections between academic research and practical maritime governance.

Key figures beyond Janiszewski include scholars who have extended and updated maritimization theory, applied it to new empirical domains, and engaged it with contemporary theoretical debates [3], [7]. While Janiszewski is identified as a principal proponent, the tradition represents a collective intellectual project sustained across decades.

2.2.6 Contemporary Relevance and Limitations

Recent scholarship has revisited Janiszewski's maritimization theory to assess its relevance for contemporary maritime and Blue Economy challenges. This reassessment reveals both enduring strengths and important limitations.

Strengths for Blue Economy analysis include:

Holistic linkage of material and social dimensions: The two-factor framework provides a heuristic for analyzing how environmental conditions, technological systems, and institutional arrangements interact with human agency, cultural practices, and social organization [1]. This holistic approach is valuable for Blue Economy analysis, which must account for both ecological-technological constraints and social-institutional mediations of marine resource use.

Attention to institutional embedding and path dependency: Janiszewski's emphasis on how maritime institutions create self-reinforcing developmental pathways illuminates the challenges of Blue Economy transitions [1], [5], [7]. Existing maritime institutional configurations—fleet ownership structures, port governance arrangements, educational systems, regulatory frameworks—shape the possibilities for and resistances to new blue growth initiatives.

Focus on cultural reproduction and identity: The framework's attention to how maritime identities and practices are transmitted across generations highlights the cultural dimensions of maritime sustainability [5]. Blue Economy initiatives that ignore existing maritime cultures and identities risk generating social resistance or failing to mobilize local knowledge and commitment.

Multi-scalar applicability: Maritimization has been productively applied from occupational communities to regional development, demonstrating analytical versatility across scales [1], [2], [4], [7]. Blue Economy governance similarly requires multi-scalar analysis linking local communities, regional economies, national policies, and global markets.

Limitations and gaps identified in contemporary assessments include:

Need for political economy integration: Reviewers call for stronger incorporation of global governance structures, transnational capital flows, and multi-scalar policy dynamics that are not fully elaborated in the original maritimization framework [3]. Contemporary maritime change is shaped by global shipping networks, international fisheries regimes, transnational corporate strategies, and supranational governance institutions that exceed the framework's original focus on national and regional scales.

Methodological updating requirements: Contemporary maritime transformations—driven by digitalization, automation, climate change, and global supply chain restructuring—require empirical tools and interdisciplinary methods beyond those employed in mid-twentieth-century field studies [3], [7]. The tradition's qualitative, community-focused methods need complementing with network analysis, global commodity chain analysis, and multi-sited ethnography to capture contemporary maritime complexity.

Limited engagement with environmental change: While Janiszewski's marine factor includes environmental conditions, the framework was developed before contemporary awareness of anthropogenic ocean change, marine biodiversity loss, and climate impacts on marine systems [3]. Blue Economy analysis requires more explicit integration of ecological dynamics, ecosystem services, and environmental limits than the original framework provides.

Insufficient attention to power asymmetries and conflict: The framework emphasizes how maritime factors and human factors interact to produce maritimization but provides limited analytical tools for examining power relations, distributional conflicts, and resistance within maritime development processes [3]. Blue Economy initiatives generate winners and losers, displace existing users, and concentrate benefits—dynamics requiring explicit power analysis.

Despite these limitations, Janiszewski's maritimization theory provides essential ontological foundations for Blue Sociology. The framework's core insight—that sea and society are mutually constitutive through ongoing material-institutional-cultural entanglements—remains vital for understanding how Blue Economy initiatives reshape coastal communities, maritime industries, and ocean governance. The theory's semi-deterministic stance, rejecting both environmental determinism and social constructivism, offers a productive middle ground for analyzing socioecological systems. And its attention to mechanisms of institutional embedding and cultural reproduction illuminates the path dependencies and social resistances that shape Blue Economy transitions.

2.2.7 Supporting the Marine-Maritime-Oceanic Tripartite Framework

Janiszewski's maritimization theory directly supports Blue Sociology's tripartite analytical distinction between Marine, Maritime, and Oceanic domains:

The Marine domain—natural oceanographic and ecological processes—corresponds to key elements of Janiszewski's marine factor, particularly environmental conditions and resource distributions that structure maritime possibilities [1]. However, Blue Sociology extends this domain by incorporating contemporary understanding of marine ecosystems, biogeochemical cycles, and anthropogenic environmental change.

The Maritime domain—human activities, labor regimes, and institutional arrangements at sea—is illuminated by Janiszewski's analysis of how the marine factor and human factor interact to produce specific forms of maritime work, organization, and governance [1], [4]. The framework's attention to occupational communities, institutional structures, and cultural practices provides analytical tools for examining maritime labor regimes, fleet organizations, and port governance.

The Oceanic domain—planetary-scale socioecological systems and governance challenges—extends Janiszewski's framework to scales and dynamics he did not explicitly address [3]. However, the theory's emphasis on multi-scalar entanglements and institutional embedding provides conceptual foundations for analyzing how local maritime communities are linked to global ocean governance regimes, transnational commodity chains, and planetary environmental change.

Janiszewski's framework thus provides ontological grounding for Blue Sociology's core claim: that understanding contemporary ocean governance requires analyzing the mutual constitution of marine environments, maritime social organization, and oceanic-scale political economy. The theory demonstrates that sea-society relations are neither purely natural nor purely social but emerge through ongoing material-institutional-cultural entanglements—a relational ontology that Blue Sociology extends to Blue Economy analysis.

2.3 Michael Poole's Programmatic Systematization: Methodological and Thematic Foundations

2.3.1 The 1981 Foundational Article and Its Significance

Michael Poole's 1981 article "Maritime sociology: towards a delimitation of themes and analytical frameworks," published in *Maritime Policy & Management*, represents a pivotal moment in the institutionalization of maritime sociology as a coherent field of inquiry [9]. While Janiszewski's work provided ontological foundations for understanding sea-society entanglement, Poole's contribution was fundamentally programmatic: he sought to delimit maritime sociology's core thematic domains, specify its analytical frameworks, and establish an agenda for systematic empirical research and theoretical development.

The article's significance lies not in presenting a comprehensive theory of maritime social relations but in performing essential intellectual work for field formation: defining boundaries, identifying core problems, proposing methodological approaches, and calling for integration with general sociological theory [9]. Poole argued that maritime sociology risked remaining a collection of scattered case studies and descriptive accounts unless scholars deliberately constructed a systematic research program organized around clearly defined themes and explicit analytical frameworks.

Poole's intervention addressed a specific problem in maritime sociology's development: the field had generated substantial empirical research on seafarers, ports, shipping companies, and maritime labor relations, but this work remained insufficiently connected to broader sociological debates and theoretical frameworks [9]. Maritime topics were often treated as specialized niches rather than sites for developing and testing general sociological concepts. Poole called for reversing this pattern: maritime sociology should use its distinctive empirical domains to contribute to general sociological understanding of work, organizations, labor markets, and industrial relations.

The article's programmatic character is evident in its structure and rhetorical strategy. Rather than presenting original empirical research or developing a novel theoretical framework, Poole systematically mapped the thematic terrain of maritime sociology, identified key analytical problems within each theme, and specified how general sociological frameworks could be productively applied to maritime contexts [9]. This mapping exercise performed crucial boundary work, establishing what counts as maritime sociology and how it relates to adjacent fields.

2.3.2 Programmatic Agenda: Systematizing Maritime Sociology as a Field

Poole's programmatic agenda rested on several key propositions about how maritime sociology should develop as a field:

Thematic delimitation: Maritime sociology requires explicit identification of its core substantive domains rather than treating "maritime" as an undifferentiated category [9]. Poole argued that the field encompasses several distinct but interrelated thematic clusters, each requiring specific analytical approaches and raising particular theoretical questions. By delimiting these themes, maritime sociology could develop specialized expertise while maintaining coherence as a field.

Analytical framework specification: Within each thematic domain, maritime sociology should explicitly specify the analytical frameworks—theoretical concepts, causal models, methodological approaches—appropriate for investigating particular problems [9]. Rather than relying on implicit or ad hoc frameworks, the field should systematically evaluate which sociological theories and methods are most productive for maritime contexts.

Integration with general sociological theory: Maritime sociology should not remain an isolated specialty but should actively engage with and contribute to general sociological debates [9]. Maritime workplaces, organizations, and labor markets provide distinctive empirical contexts for testing and refining sociological theories of work, organizations, stratification, and industrial relations. Conversely, general sociological frameworks provide analytical tools for understanding maritime social phenomena.

Methodological pluralism: Maritime sociology requires diverse methodological approaches appropriate to different research questions and empirical contexts [9]. Qualitative ethnography, survey research, organizational case studies, historical analysis, and comparative methods all have roles in building cumulative knowledge about maritime social relations.

Empirical research agenda: The field should develop a coordinated empirical research agenda that systematically investigates core themes rather than pursuing scattered case studies [9]. This requires identifying key empirical questions, establishing research priorities, and building on prior work to generate cumulative findings.

Interdisciplinary engagement: While maintaining sociological identity, maritime sociology should engage productively with maritime economics, law, policy studies, and industry expertise [9]. Many maritime problems require integrated analysis that combines sociological insights with economic, legal, and technical knowledge.

This programmatic vision positioned maritime sociology as a legitimate sociological specialty with distinctive empirical domains, systematic analytical frameworks, and contributions to general sociological knowledge. Poole's agenda sought to move the field beyond descriptive studies of maritime occupations toward theoretically informed, methodologically rigorous, and cumulatively organized research.

2.3.3 Thematic Delimitation: Core Domains of Maritime Sociology

Poole's most influential contribution was his systematic delimitation of maritime sociology's core thematic domains. While the supplied sources do not provide exhaustive detail on all themes Poole identified, they document several key clusters that have structured subsequent maritime sociology research:

Maritime labor and employment relations constitutes a central thematic domain encompassing the organization, conditions, and governance of work at sea [9], [10], [11], [12]. This theme includes analysis of labor market structures, employment contracts, wage determination, working conditions, occupational health and safety, and labor market segmentation. Maritime labor markets exhibit distinctive features—global recruitment, flag-of-convenience regimes, multinational crews, extended periods away from home—that make them valuable sites for examining how globalization reshapes employment relations [10], [11].

Empirical research in this domain has examined global labor restructuring in shipping, analyzing how flag-of-convenience systems enable shipowners to access low-wage labor markets while evading national labor regulations [10], [11]. Studies

have documented the emergence of transnational labor supply chains connecting seafarer-sending countries (particularly the Philippines, India, and Eastern Europe) with ship operators through complex intermediary networks of manning agencies, training institutions, and union organizations [11], [13]. This research reveals how maritime labor markets are structured by intersecting national regulations, international conventions, corporate strategies, and union campaigns [10], [11].

Seafarers' occupational life and social conditions represents a second major theme focusing on the lived experiences, social relations, and welfare of maritime workers [9], [14]. This domain examines how the distinctive conditions of seafaring work —prolonged separation from families and communities, confinement in shipboard social environments, exposure to occupational hazards, social isolation—shape seafarers' identities, relationships, and wellbeing [14].

Research in this area has employed qualitative methods—participant observation, in-depth interviews, ethnography—to capture the subjective dimensions of seafaring life [14]. Studies have documented the causes and effects of seafarers' social isolation, examining how limited shore leave, communication constraints, and cultural-linguistic diversity within multinational crews create psychological stressors and mental health risks [14]. This research has informed policy debates about seafarer welfare, shore-based support services, and shipboard social organization [14].

Organizational structures and management in shipping constitutes a third thematic cluster examining how shipping companies, fleets, and vessels are organized and managed [9], [15]. This domain applies organizational sociology frameworks to analyze bureaucratic structures, management systems, authority relations, and organizational cultures in maritime contexts [15].

Empirical work has included comparative studies of shipboard organization, examining how different organizational forms —bureaucratic versus organic structures, hierarchical versus participative management—affect crew satisfaction, role innovation, and operational effectiveness [15]. Research has analyzed how shipping companies adapt organizational structures to competitive pressures, regulatory requirements, and technological change [15]. This work contributes to general organizational theory by examining how extreme environments, safety-critical operations, and multinational workforces shape organizational design.

Port labor and industrial relations represents a fourth major theme analyzing labor organization, industrial conflict, and collective bargaining in port contexts [9], [16]. Ports have historically been sites of strong labor organization, militant unionism, and significant industrial conflict, making them valuable contexts for examining labor movement strategies, employer responses, and state labor regulation [16].

Research has examined the historical evolution of dock labor systems, comparing different national models of port labor organization and their transformations under pressures of containerization, privatization, and global competition [16]. Studies have analyzed how port unions have responded to technological change, employment casualization, and employer strategies to weaken labor organization [16]. This research illuminates broader dynamics of labor movement adaptation to economic restructuring.

Transnational union organization and regulation emerges as a fifth theme examining how maritime unions have developed transnational strategies to regulate global shipping [9], [10], [11]. The International Transport Workers' Federation (ITF) flag-of-convenience campaign represents a prominent case of transnational union coordination, attempting to establish minimum standards for wages and working conditions on ships registered in open registries [10], [11].

Research has analyzed the ITF's organizational structure, bargaining strategies, and effectiveness in regulating flag-of-convenience shipping [10], [11]. Studies have examined tensions between national unions' interests and international solidarity, the role of seafarer-sending countries' unions in transnational campaigns, and the conditions under which transnational union coordination succeeds or fails [10], [11]. This work contributes to understanding of transnational labor politics and global labor governance.

These thematic domains are not exhaustive but represent core areas that Poole identified as central to maritime sociology's research agenda. Each theme encompasses specific empirical questions, analytical challenges, and theoretical debates that have structured subsequent research [9].

2.3.4 Linking Maritime Studies to General Sociological Theory

A central element of Poole's programmatic vision was the insistence that maritime sociology should systematically link its substantive topics to general sociological theory rather than treating maritime phenomena as sui generis [9]. This linkage serves dual purposes: it enables maritime sociology to draw on established theoretical frameworks to analyze maritime social relations, and it positions maritime empirical research as contributing to general sociological knowledge.

Labor market theory provides frameworks for analyzing maritime employment relations, wage determination, and labor market segmentation [9], [10], [11]. Maritime labor markets exhibit extreme forms of globalization, with workers recruited internationally, employed on vessels registered in third countries, and governed by complex intersections of national and international regulations. Applying labor market theory to maritime contexts illuminates how global labor markets are structured, how workers and employers navigate regulatory complexity, and how labor market institutions shape distributional outcomes [10], [11].

Organizational theory offers concepts for analyzing shipping companies, fleets, and shipboard organization [9], [15]. Maritime organizations operate in distinctive environments—geographically dispersed operations, safety-critical activities, multinational workforces, complex regulatory frameworks—that create specific organizational challenges. Applying organizational theory to maritime contexts tests and refines concepts of organizational structure, management systems, and organizational culture under extreme conditions [15].

Industrial relations theory provides frameworks for analyzing collective bargaining, labor-management relations, and union strategies in maritime contexts [9], [10], [11], [16]. Maritime industrial relations exhibit distinctive features—transnational union coordination, flag-of-convenience systems, multinational crews—that challenge conventional industrial relations

frameworks developed for national labor markets. Maritime research thus contributes to understanding of how industrial relations systems adapt to globalization [10], [11].

Occupational sociology offers concepts for analyzing seafaring as an occupation, including occupational identity, socialization, career patterns, and occupational communities [9], [14]. Seafaring represents an extreme case of occupational distinctiveness—prolonged separation from shore communities, confinement in total institutions, exposure to significant hazards—making it valuable for examining how occupational conditions shape identity and social relations [14].

Stratification theory provides frameworks for analyzing inequality, mobility, and stratification within maritime labor markets [9], [11], [13]. Maritime labor markets are characterized by significant stratification along national, racial, and skill dimensions, with officers predominantly from developed countries and ratings from developing countries. Analyzing maritime stratification illuminates how global inequality is reproduced through labor market institutions [11], [13].

By systematically applying these general theoretical frameworks to maritime contexts, maritime sociology both gains analytical power and contributes to theoretical development. Maritime empirical research provides distinctive cases for testing theoretical propositions, revealing scope conditions, and generating theoretical refinements [9].

2.3.5 Methodological Pluralism and Empirical Research Agenda

Poole's programmatic vision embraced methodological pluralism, recognizing that different research questions and empirical contexts require different methodological approaches [9]. The subsequent development of maritime sociology has exemplified this pluralism, employing diverse methods to investigate maritime social relations.

Qualitative ethnography and participant observation have been extensively used to examine seafarers' lived experiences, shipboard social relations, and occupational cultures [14]. These methods enable researchers to capture subjective dimensions of maritime work, observe social interactions in situ, and understand how seafarers interpret and navigate their occupational conditions. Studies of seafarer social isolation, welfare needs, and mental health have relied heavily on qualitative methods to access workers' perspectives and experiences [14].

Survey research and quantitative analysis have been employed to examine labor market patterns, employment conditions, and workforce characteristics across large samples [15]. Surveys of seafarers, shipping companies, and maritime organizations provide data on wages, working conditions, career patterns, and organizational practices that enable statistical analysis of maritime labor markets and organizations [15].

Comparative case studies have been used to analyze organizational structures, management systems, and industrial relations across different shipping companies, fleets, or national contexts [15], [16]. Comparative methods enable identification of how organizational and institutional variations shape maritime social relations and outcomes [15], [16].

Historical analysis has examined the evolution of maritime labor systems, union organization, and regulatory regimes over time [16]. Historical research illuminates how contemporary maritime institutions emerged, how they have been transformed by technological and economic change, and how past struggles shape present possibilities [16].

Network analysis and institutional mapping have been applied to examine transnational union coordination, labor supply chains, and regulatory networks [10], [11]. These methods reveal the complex organizational structures and inter-organizational relationships that constitute global maritime governance [10], [11].

Policy analysis and document review have been used to examine regulatory frameworks, international conventions, and industry standards that govern maritime labor and operations [10], [11]. Policy analysis connects empirical research on maritime social relations to governance debates and reform proposals [10], [11].

This methodological pluralism reflects Poole's vision of maritime sociology as a field that adapts methods to research questions rather than privileging particular methodological approaches [9]. The diversity of methods employed in maritime sociology research demonstrates the field's maturity and analytical sophistication.

2.3.6 Influence on Subsequent Maritime Sociology Scholarship

While the supplied sources do not provide systematic historiography of Poole's direct influence on subsequent scholarship, they document research programs that exemplify his programmatic vision. The thematic domains Poole identified—maritime labor, seafarers' occupational life, organizational structures, port labor, industrial relations—have indeed structured subsequent empirical research [10], [11], [14], [15], [16].

Transnational labor governance research has developed as a major research program examining how maritime unions, international organizations, and regulatory regimes attempt to govern global shipping labor markets [10], [11], [13]. This work directly addresses themes Poole identified as central to maritime sociology, applying industrial relations and labor market theory to analyze transnational union strategies, flag-of-convenience regulation, and international labor standards [10], [11].

Seafarer welfare and social conditions research has emerged as a sustained research program examining the causes and consequences of seafarers' social isolation, mental health challenges, and welfare needs [14]. This research employs the qualitative methods and occupational sociology frameworks that Poole advocated for understanding seafarers' lived experiences [14].

Maritime organizational research has examined how shipping companies adapt organizational structures and management systems to competitive pressures, regulatory requirements, and technological change [15]. This work applies organizational theory to maritime contexts in the manner Poole proposed [15].

Port labor and industrial relations research has analyzed the transformation of dock labor systems under pressures of containerization, privatization, and globalization [16]. This research examines union strategies, employer responses, and regulatory changes in port contexts, addressing themes Poole identified as central to maritime sociology [16].

Global labor market research has examined how maritime labor markets are structured by intersecting national regulations, international conventions, corporate strategies, and union campaigns [10], [11], [13]. This work analyzes maritime labor markets as cases of global labor market dynamics, contributing to general understanding of how globalization reshapes employment relations [10], [11].

These research programs demonstrate that Poole's programmatic vision has been substantially realized: maritime sociology has developed as a field with clearly delimited thematic domains, systematic application of general sociological theory, methodological pluralism, and cumulative empirical research. While individual scholars may not explicitly cite Poole's 1981 article, the field's development reflects his programmatic agenda.

2.3.7 Relevance for Analyzing Blue Economy Labor Regimes and Institutional Power

Poole's programmatic systematization provides essential methodological and thematic foundations for Blue Sociology's analysis of Blue Economy labor regimes and institutional power. Several elements of his framework are particularly relevant:

Thematic organization provides a template for systematically analyzing Blue Economy sectors. Just as Poole delimited core themes in maritime sociology—labor markets, occupational life, organizational structures, industrial relations—Blue Sociology can delimit core analytical dimensions across Blue Economy sectors: labor regimes, institutional arrangements, distributional outcomes, environmental impacts, governance mechanisms, and legitimacy dynamics. This thematic organization enables systematic comparison across sectors and identification of common patterns.

Integration with general theory supports Blue Sociology's ambition to contribute to broader sociological debates about labor, organizations, governance, and sustainability transitions. By explicitly linking Blue Economy analysis to labor market theory, organizational theory, political economy, and environmental sociology, Blue Sociology can both draw on established frameworks and generate insights relevant beyond maritime contexts.

Methodological pluralism legitimates Blue Sociology's use of diverse methods—ethnography, surveys, case studies, policy analysis, network analysis—appropriate to different research questions and empirical contexts. Blue Economy analysis requires methods capable of capturing lived experiences of maritime workers, organizational dynamics of blue growth firms, policy processes in ocean governance, and transnational networks of capital and regulation.

Labor-centered analysis reflects Poole's emphasis on maritime labor as a central analytical focus. Blue Sociology extends this labor-centered approach across Blue Economy sectors, examining how blue growth initiatives reshape employment relations, working conditions, occupational identities, and labor organization in fisheries, aquaculture, shipping, ports, offshore energy, and marine tourism.

Institutional analysis builds on Poole's attention to organizational structures, industrial relations, and regulatory frameworks. Blue Sociology examines how Blue Economy initiatives are shaped by and reshape institutional arrangements: property regimes, regulatory frameworks, governance organizations, industry associations, and labor unions. Understanding institutional power—who controls key resources, who participates in decision-making, whose interests are privileged—is central to Blue Economy analysis.

Transnational dynamics extend Poole's recognition that maritime phenomena increasingly require transnational analysis. Blue Economy governance involves complex interactions between local communities, national governments, regional organizations (like the EU), international conventions, and transnational corporations. Blue Sociology must analyze how these multi-scalar institutional arrangements shape distributional outcomes and governance legitimacy.

2.3.8 Supporting the Maritime Domain Analysis

Poole's framework directly supports Blue Sociology's analysis of the Maritime domain—human activities, labor regimes, and institutional arrangements at sea. His thematic delimitation provides a systematic template for examining maritime social organization:

Labor regimes: Poole's emphasis on maritime labor markets, employment relations, and working conditions provides analytical foundations for examining how Blue Economy initiatives reshape maritime work. Blue Sociology extends this analysis across sectors, comparing labor regimes in fishing, shipping, offshore energy, and marine tourism to identify common patterns and sector-specific dynamics.

Organizational structures: Poole's attention to shipping companies, fleets, and shipboard organization informs Blue Sociology's analysis of how blue growth firms are organized and managed. This includes examining corporate structures, ownership patterns, management systems, and organizational cultures in Blue Economy sectors.

Industrial relations: Poole's focus on collective bargaining, union organization, and labor-management relations supports Blue Sociology's analysis of how maritime workers organize collectively, how unions respond to Blue Economy transformations, and how industrial relations institutions mediate labor-capital conflicts.

Occupational communities: Poole's recognition of seafarers as a distinctive occupational community informs Blue Sociology's analysis of how Blue Economy initiatives affect maritime occupational identities, career patterns, and social relations. This includes examining how technological change, regulatory shifts, and economic restructuring reshape what it means to be a fisher, seafarer, or offshore worker.

Institutional power: Poole's programmatic vision implicitly recognizes that maritime social relations are shaped by institutional arrangements and power relations. Blue Sociology makes this explicit, examining how institutional power—control over resources, participation in governance, influence over policy—shapes Blue Economy outcomes and legitimacy.

Poole's systematization thus provides methodological and thematic foundations for Blue Sociology's Maritime domain analysis, enabling systematic examination of how human activities, labor regimes, and institutional arrangements at sea are transformed by Blue Economy initiatives.

2.4 Synthesis and Integration: Complementary Foundations for Blue Sociology

Janiszewski's maritimization theory and Poole's programmatic systematization provide complementary foundations for Blue Sociology, each addressing different but equally essential dimensions of the framework's theoretical architecture.

2.4.1 Complementary Contributions

Janiszewski provides ontological foundations for understanding sea-society relations as mutually constitutive entanglements. His two-factor framework—marine factor and human factor—offers a relational ontology that rejects both environmental determinism and social constructivism. This ontological stance is essential for Blue Sociology's analysis of how marine environments, maritime activities, and oceanic governance are co-produced through material-institutional-cultural processes [1]. Janiszewski's emphasis on mechanisms of entanglement—economic linkage, institutional embedding, cultural transmission, spatial concentration—provides conceptual tools for analyzing how Blue Economy initiatives become embedded in coastal communities and maritime sectors [1], [5].

Poole provides methodological and thematic systematization for organizing maritime sociology as a coherent field of inquiry. His delimitation of core themes—labor markets, occupational life, organizational structures, industrial relations—and his insistence on linking maritime topics to general sociological theory provide a template for systematic empirical research and theoretical development [9]. This systematization is essential for Blue Sociology's ambition to develop as a rigorous analytical framework rather than a collection of scattered case studies. Poole's methodological pluralism legitimates diverse research approaches appropriate to different Blue Economy questions and contexts [9].

Together, they ground Blue Sociology's dual focus on material conditions and social organization. Janiszewski's framework emphasizes how material marine conditions—environmental dynamics, technological systems, infrastructural configurations—shape social possibilities while being mediated by human agency and institutional choices [1]. Poole's framework emphasizes how social organization—labor markets, organizational structures, industrial relations, occupational communities—structures maritime activities and outcomes [9]. Blue Sociology integrates these emphases, analyzing how Blue Economy initiatives are shaped by both material marine conditions and social-institutional arrangements.

2.4.2 Informing the Six Core Analytical Dimensions

The synthesis of Janiszewski and Poole informs Blue Sociology's six core analytical dimensions:

1. Multi-scalar socioecological integration draws on Janiszewski's emphasis on how marine environments and human societies are entangled across scales—from occupational communities to regional development [1], [2], [4], [7]. It extends this by incorporating Poole's recognition that maritime phenomena increasingly require transnational analysis [9], [10], [11]. Blue Sociology analyzes how Blue Economy initiatives link local marine ecosystems, regional maritime economies, national ocean policies, and global governance regimes.

2. Labor-centered social analysis directly reflects Poole's programmatic emphasis on maritime labor as a central analytical focus [9], [10], [11], [14]. It incorporates Janiszewski's attention to how maritime occupational communities are constituted through the interplay of marine conditions and human practices [1], [4]. Blue Sociology examines labor regimes, working conditions, occupational identities, and labor organization across Blue Economy sectors.

3. Institutional power and governance builds on both frameworks' attention to institutions. Janiszewski emphasizes institutional embedding as a mechanism of maritimization [1], [5], while Poole emphasizes organizational structures and industrial relations as core themes [9], [15], [16]. Blue Sociology analyzes how institutional arrangements—property regimes, regulatory frameworks, governance organizations—shape Blue Economy outcomes and distribute power.

4. Distributional justice and equity extends both frameworks' implicit concern with how maritime development affects different groups. Janiszewski's analysis of fishing communities and port cities reveals how maritimization creates winners and losers [1], [4], [7], while Poole's emphasis on labor markets and industrial relations highlights distributional conflicts [9], [10], [11]. Blue Sociology makes distributional analysis explicit, examining who benefits and who bears costs from Blue Economy initiatives.

5. More-than-human and infrastructural agencies extends Janiszewski's marine factor to incorporate contemporary understanding of marine ecosystems, non-human species, and technological infrastructures as active agents [1]. While neither Janiszewski nor Poole explicitly developed more-than-human frameworks, Janiszewski's emphasis on material marine conditions provides foundations for analyzing how non-human agencies shape maritime social relations [1].

6. Democratic and place-based governance builds on both frameworks' attention to local contexts and institutional arrangements. Janiszewski's analysis of coastal communities emphasizes how local capacity to organize institutions and cultural practices shapes maritime development outcomes [1], [2], [5], while Poole's emphasis on industrial relations highlights the importance of worker participation and collective organization [9], [10], [11]. Blue Sociology extends these insights to argue for democratic, participatory governance mechanisms in Blue Economy transitions.

2.4.3 Theoretical Renewal and Extension

While Janiszewski and Poole provide essential foundations, Blue Sociology extends their frameworks in several ways to address contemporary Blue Economy challenges:

Explicit environmental integration: Blue Sociology incorporates contemporary understanding of marine ecosystems, biodiversity, climate change, and planetary boundaries that were not central to either framework [3]. This requires analyzing how Blue Economy initiatives affect and are affected by marine environmental change.

Global political economy: Blue Sociology explicitly analyzes transnational capital flows, global commodity chains, and multi-scalar governance that exceed the national and regional scales emphasized in both frameworks [3], [10], [11]. This requires examining how Blue Economy initiatives are shaped by global economic structures and power relations.

Justice and equity frameworks: Blue Sociology makes distributional justice, procedural equity, and recognition explicit analytical priorities, drawing on environmental justice and blue justice scholarship [3]. This requires examining not just how Blue Economy initiatives affect different groups but whether these effects are just and legitimate.

More-than-human ontologies: Blue Sociology incorporates insights from oceanic turn scholarship on hydrosocial relations, multispecies entanglements, and infrastructural politics [3]. This requires analyzing how non-human agencies—marine species, ocean currents, technological systems—actively shape Blue Economy outcomes.

Sustainability transitions: Blue Sociology frames Blue Economy development as a sustainability transition challenge, requiring analysis of how existing maritime systems can be transformed toward more socially and ecologically sustainable configurations [3]. This requires examining path dependencies, transition pathways, and transformation strategies.

These extensions do not reject Janiszewski and Poole's frameworks but build on their foundations to address challenges they did not explicitly confront. The result is a Blue Sociology framework that maintains continuity with maritime sociology traditions while incorporating contemporary theoretical innovations and addressing twenty-first-century ocean governance challenges.

2.5 Conclusion: Theoretical Foundations for Blue Sociology

Ludwik Janiszewski's maritimization theory and Michael Poole's programmatic systematization provide complementary and essential foundations for Blue Sociology. Janiszewski offers an ontological framework for understanding how marine environments and human societies become mutually constitutive through processes of economic linkage, institutional embedding, cultural transmission, and spatial concentration. His semi-deterministic approach—recognizing both material constraints and human agency—provides conceptual tools for analyzing how Blue Economy initiatives reshape sea-society relations while being shaped by existing maritime institutions and cultures.

Poole offers methodological and thematic systematization that organizes maritime sociology as a coherent field with clearly delimited domains, systematic application of general sociological theory, and methodological pluralism. His programmatic vision—linking maritime labor, occupational life, organizational structures, and industrial relations to broader sociological debates—provides a template for developing Blue Sociology as a rigorous analytical framework capable of contributing to general sociological knowledge while addressing specific Blue Economy challenges.

Together, these frameworks ground Blue Sociology's dual focus on material sea-society entanglements and social organization of maritime labor, institutions, and power. They inform the framework's six core analytical dimensions and support its tripartite distinction between Marine, Maritime, and Oceanic domains. While both frameworks require extension to address contemporary challenges—global political economy, environmental change, justice concerns, more-than-human agencies—they provide essential continuity with maritime sociology traditions and demonstrate that understanding Blue Economy requires sustained engagement with the field's theoretical and empirical foundations.

The synthesis reveals that Blue Sociology is not a radical departure from maritime sociology but a complementary renewal that extends the field's analytical tools to address accelerating Blue Economy development, intensifying ocean governance challenges, and urgent social sustainability imperatives. By grounding Blue Sociology in Janiszewski's ontological insights and Poole's programmatic systematization, we ensure that the framework maintains analytical rigor, empirical grounding, and theoretical coherence while addressing twenty-first-century ocean governance challenges.

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Integration Note: This expanded section is designed to replace or substantially augment Section 2 ("Theoretical Foundations: From Maritime Sociology to Blue Sociology") in the main Blue Sociology Framework Synthesis document. The section maintains the document's academic tone, citation style (IEEE numeric), and analytical rigor while providing comprehensive coverage of Janiszewski's maritimization theory and Poole's programmatic systematization as foundational contributions to Blue Sociology.